

# Resolving a Conjecture on Stack-Sorting

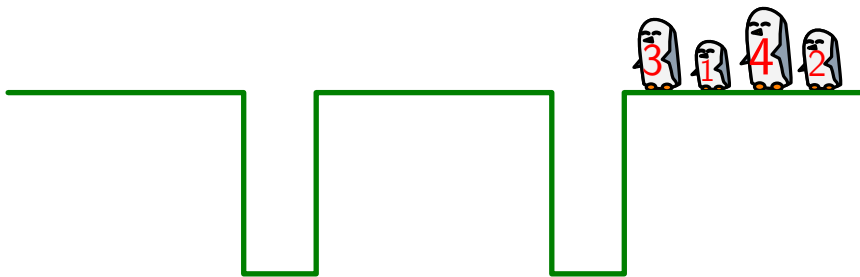
Michael Luo  
Research Science Institute  
Mentor: Ryota Inagaki

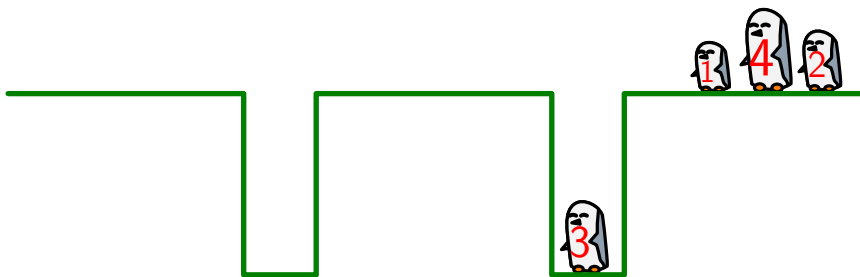
August 2026

# Penguins

Michael Luo  
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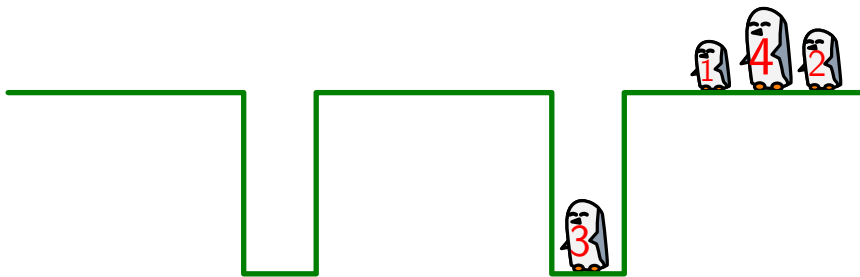
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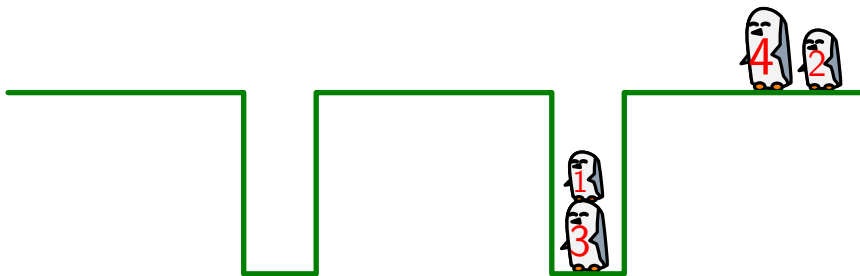
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- **Impatient:** small penguins will always jump on big penguins



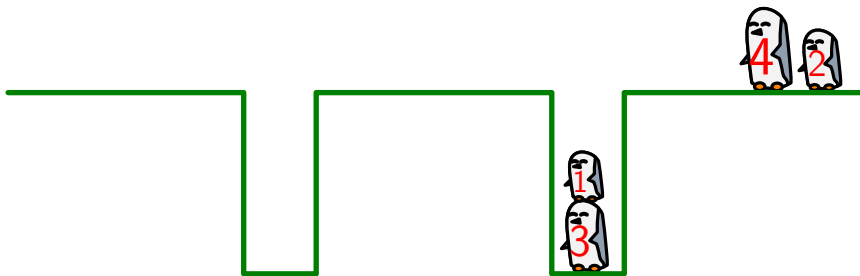
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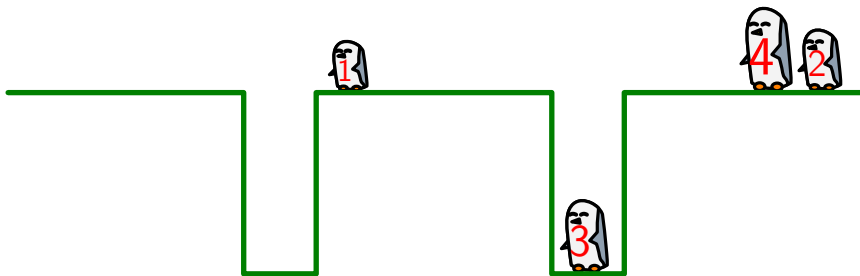
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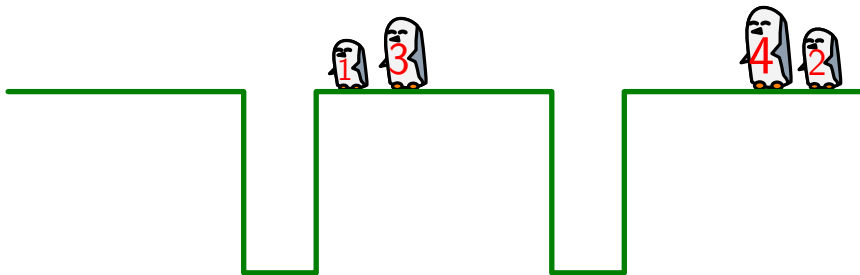
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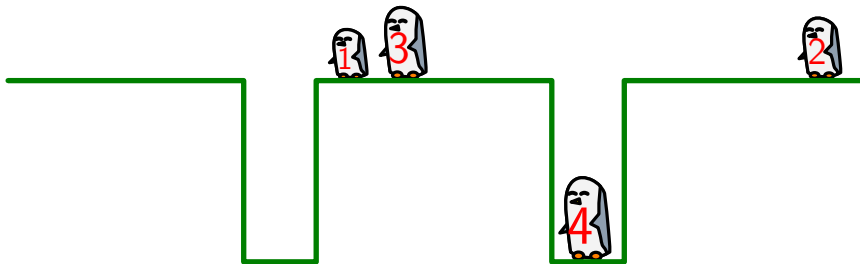
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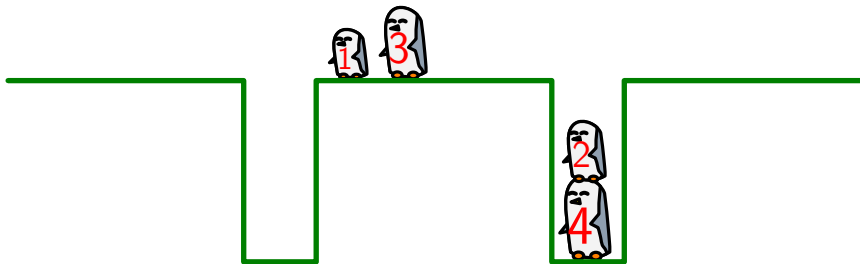
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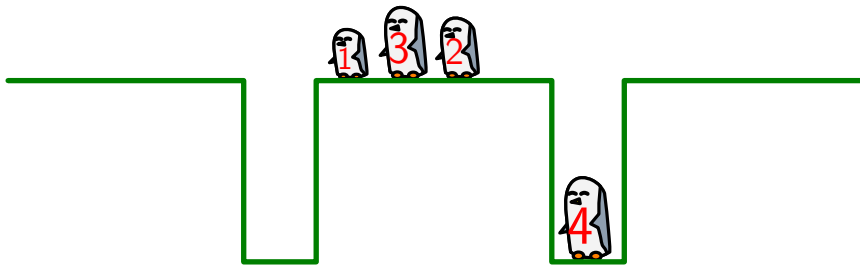
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Penguins by Evan Chen, used with permission. Figure generated with AI assistance.

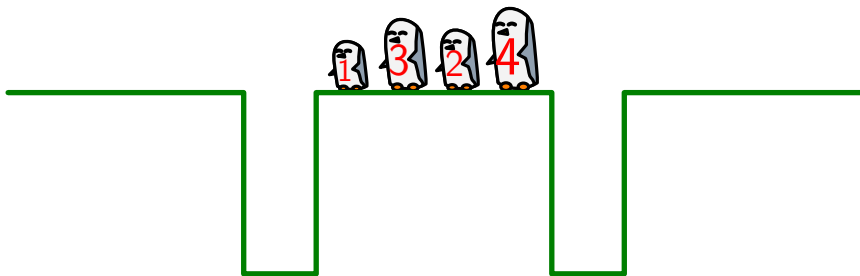
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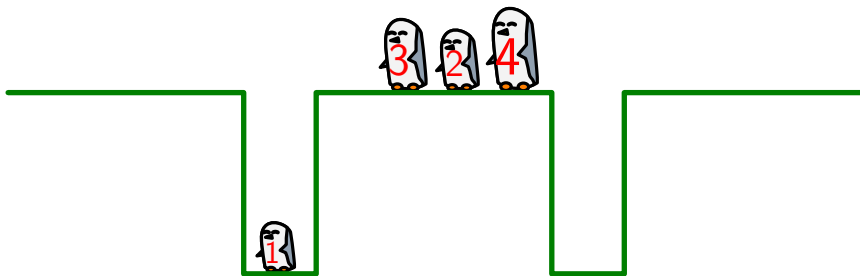
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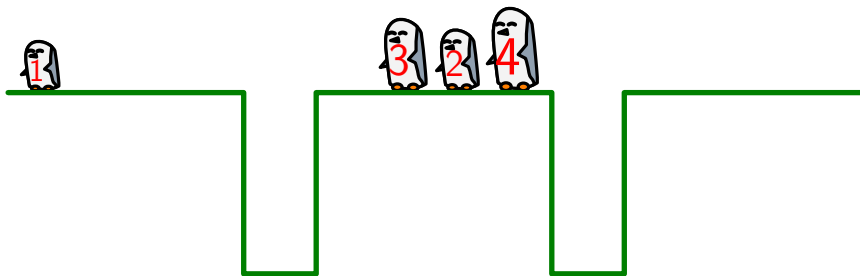
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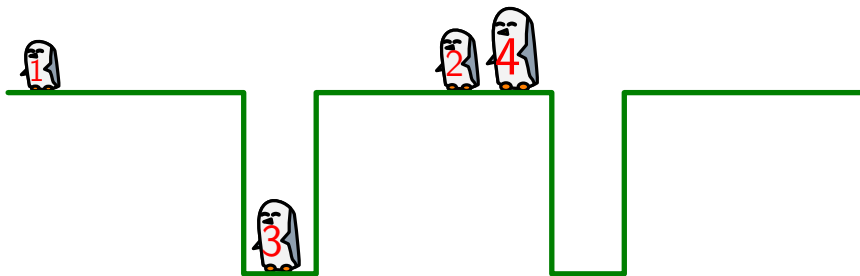
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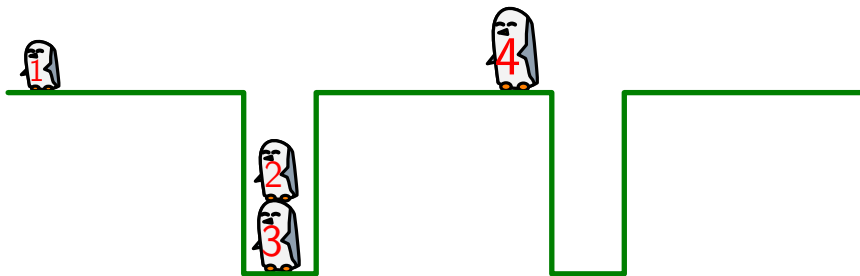
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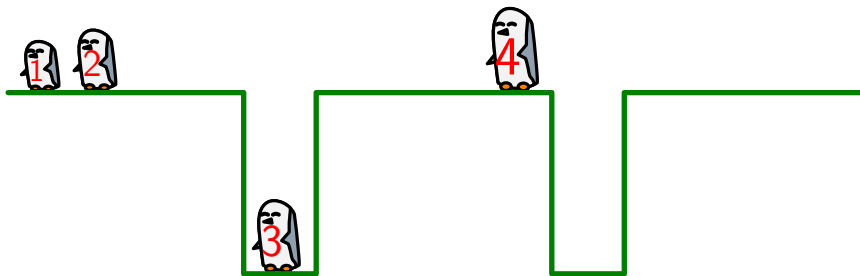
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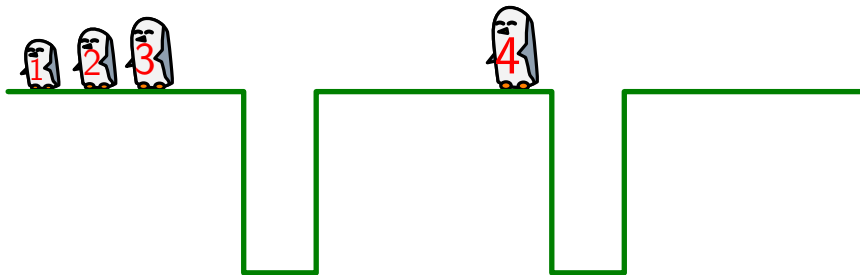
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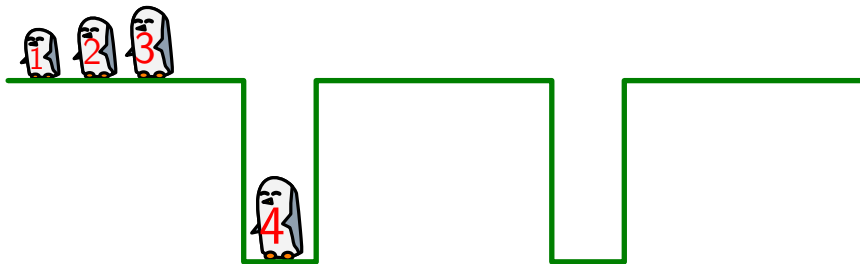
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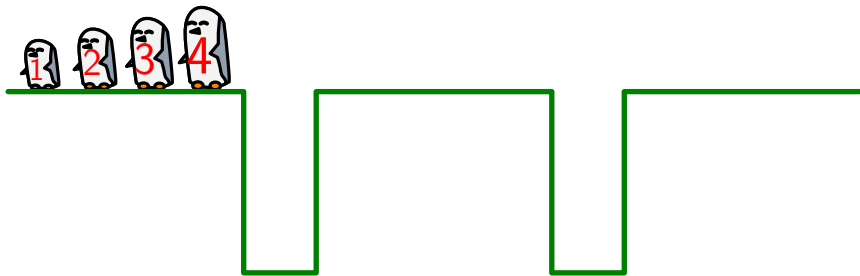
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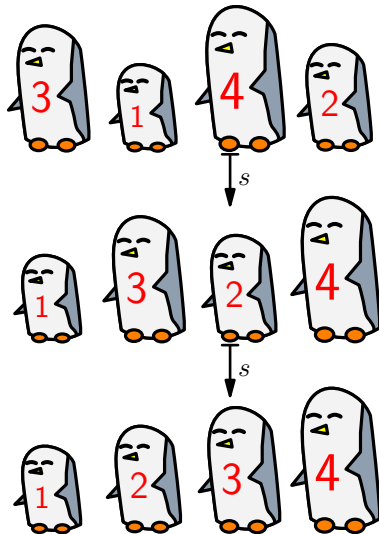


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In notation:  $s(3142) = 1324$ ,  $s(1324) = 1234$

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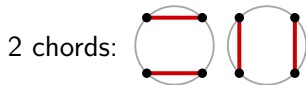
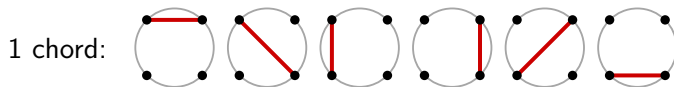
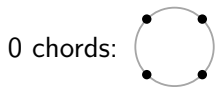
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Theorem (L. 2026)

*# of  $s^2$ -sortable permutations in  $S'_n$  is  $n$ th Motzkin number.*

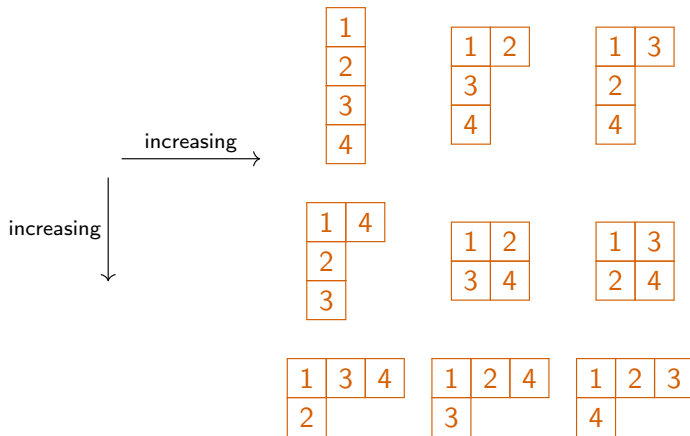
$n$ th Motzkin counts:

Ways to draw any # of disjoint chords connecting  $n$  points on a circle



$n$ th Motzkin counts:

Number of **standard Young tableaux** with width  $\leq 3$  (Regev 1981)



nth Motzkin counts:

Number of 4321-, (3412,2413)-, (3412,3241) and 3412-avoiding involutions in  $S_n$ .  
Number of sequences of length  $n-1$  consisting of positive integers such that the first and last elements are 1 or 2 and the absolute difference between any two consecutive elements is 0 or 1. - [Jon Perry](#), Sep 04 2003  
From [David Callan](#), Jul 15 2004: (Start)  
Also number of Motzkin  $n$ -paths: paths from  $(0,0)$  to  $(n,0)$  in an  $X \times Y$  grid using only steps  $U = (1,1)$ ,  $F = (1,0)$  and  $D = (1,-1)$ .  
Number of Dyck  $n$ -paths with no UUU. (Given such a Dyck  $n$ -path, change each UUU to  $UDU$  and then change each remaining UUD to  $F$ . This is a bijection to Motzkin  $n$ -paths. Example with  $n=5$ :  $UUDUUDUDD \rightarrow UUFUDDU$ .)  
Number of Dyck  $(n+1)$ -paths with no UUU. (Given such a Dyck  $(n+1)$ -path, mark each  $U$  that is followed by a  $D$  and each  $D$  that is not followed by a  $U$ . Then change each unmarked  $U$  and each matching  $D$  to  $F$ . Lastly, delete all the marked steps. This is a bijection to Motzkin  $n$ -paths. Example with  $n=6$  and marked steps in small type:  $UUDUUDUDDU \rightarrow UUDUUDUDDU \rightarrow UUDUUDUDDU \rightarrow UUDUUDUDDU$ . [Erf]  $a(n)$  is the number of strings of length  $2n+2$  from the following recursively defined set:  $L$  contains the empty string and, for any strings  $a$  and  $b$  in  $L$ , we also find  $(ab)$  in  $L$ . The first few elements of  $L$  are  $e$ ,  $(1,0)$ ,  $((1,1))$ ,  $((1,0),(1,0))$ ,  $((1,1),(1,1))$ ,  $((1,0),(1,0),(1,0))$  and so on. The primes that  $a(n)$  is less than or equal to  $C(n)$ , the  $n$ -th Catalan number. See [Orinick link](#) (2024). - [Saul Schleier](#) [[Isaacs\(AT\)math.rutgers.edu](#), Feb 23 2006] (Additional Linked content added by [William P. Orrick](#), Jun 13 2024).  
 $a(n)$  = number of Dyck  $n$ -paths all of whose valleys have even  $x$ -coordinate (when path starts at origin). For example,  $T(4,2)=3$  counts  $UUUUUUUU$ ,  $UUUUUUUU$ ,  $UUUUUUUU$ . Given such a path, split it into  $n$  subpaths of length 2. Each subpath is of type  $UD$  or  $DU$  (there will be no  $DU$  for that would entail a valley with odd  $x$ -coordinate). This is a bijection to Motzkin  $n$ -paths. - [David Callan](#), Jun 07 2006  
Also the number of standard Young tableaux of height  $\leq 3$ . - [Mike Zabrocki](#), Jun 24 2007  
 $a(n)$  is the number of RNA shapes of size  $2n+2$ . RNA shapes are essentially Dyck words without "directly nested" motifs of the form  $A1B1B1C$ , for  $A, B$  and  $C$  Dyck words. The first RNA shapes are  $\{(); (1,1); ((1,1)); ((1,1),(1,1)); ((1,1),(1,1),(1,1)); \dots\}$ . - [Yann Ponty](#) [[ponty\(AT\)iri.fr](#)], May 30 2007  
The sequence is self-generated from top row  $A$  going to the left starting  $(1,1)$  and bottom row  $B$ , the same sequence but starting  $(0,1)$  and going to the right. Take dot product of  $A$  and  $B$  and add the result to  $n$ -th term of  $A$  to get the  $(n+1)$ -th term of  $A$ . Example:  $a(5) = 21$  as follows: Take dot product of  $A = (0, 4, 2, 1, 1)$  and  $B = (1, 1, 2, 4, 0)$ ,  $0 + 1 = 1$ ,  $4 + 2 = 6$ ,  $2 + 4 = 6$ ,  $1 + 0 = 1$ ,  $1 + 0 = 1$ .  
Equals  $A085773 + A085737$  shifted (i.e.,  $(1,2,3,13,35,96,\dots) \rightarrow (1,1,2,5,13,35,96,\dots)$ ). - [Gary W. Adamson](#), Dec 28 2008  
Starting with offset 1 iterates of  $M = (1,1,8,0,\dots)$ , where  $M$  is a triangular matrix with  $(0,1,1,1,\dots)$  in the main diagonal and  $(1,1,1,\dots)$  in the super and subdiagonals. - [Gary W. Adamson](#), Jan 07 2009  
 $a(n)$  is the number of involutions of  $\{1,2,\dots,n\}$  having genus 0. The genus  $g(p)$  of a permutation  $p$  of  $\{1,2,\dots,n\}$  is defined by  $g(p)=1/2(n(2-p)-\text{cyc}(p))$ , where  $p$  is the inverse permutation of  $p$ ,  $c$  is  $\text{cyc}(p)$ ,  $\text{cyc}(p)$  is the number of cycles of the permutation  $p$ . Example:  $a(4)=9$ . Indeed,  $p=3412=(13)(24)$  is the only involution of  $\{1,2,3,4\}$  with genus  $x$ . This follows easily from the fact that a permutation  $p$  of  $\{1,2,\dots,n\}$  has genus 0 if and only if the cycle decomposition of  $p$  gives a noncrossing partition of  $\{1,2,\dots,n\}$  and each cycle of  $p$  is increasing (see Lemma 2.1 of the [Bulucea-Sandion reference](#)). [Also, redundantly, for  $p=3412=(13)(24)$  we have  $c(p)=2341=3412=4123=(1432)$  and so  $g(p)=(1/2)(4+1-1)=1$ . - [Emeric Deutsch](#), May 21 2018  
Let  $w_i,j,n$  denote walks in  $\mathbb{N}^2$  which satisfy the multivariate recurrence  $w_{i+1,1,n} = w_{i,1,n} + w_{i-1,n} + w_{i-1,n-1} + w_{i-1,n-1} + w_{i-1,n-1} + w_{i-1,n-1}$  with boundary conditions  $w(0,0,1) = 1$  and  $w_{i,j,n} = 0$  if  $i < 0$  or  $j < 0$  or  $i < 0$ . Then  $a(n) = \sum_{i=0}^n w_{i,0,n} = \sum_{i=0}^n w_{i,n,n}$ , the number of such walks of length  $n$ . - [Peter Luschny](#), May 21 2018  
 $a(n)/n!$  tends to  $3/8$  as  $N \rightarrow \infty$ :  $(1+2\cos(2\pi/3))$  relating to Longest odd  $N$  regular polygon diagonals, by way of example,  $n=7$ : Using the tridagonal generator [cf. comment of Jan 07 2009], for  $n=7$  (from  $n=7$ ), we extract  $n=7$   $3 \times 3$  matrix,  $(0,1,0,1,1; 0,1,1; 0,1,1)$  with an eval of 2.24697... the longest Heptagon diagonal with edge  $= 1$ . As  $N$  tends to infinity, the diagonal lengths tend to  $3/8$ , the convergent of the sequence. - [Gary W. Adamson](#), Jun 08 2011  
Number of  $(n+1)$ -permutations avoiding the pattern 132 and the dotted pattern 23'dot(1). - [Jean-Luc Baril](#), Mar 07 2012  
Number of  $n$ -length words  $w$  over alphabet  $\{a,b,c\}$  such that for every prefix  $x$  of  $w$  have  $\#(x,a) \geq \#(x,b)$ ,  $\#(x,b) \geq \#(x,c)$ , where  $\#(x,i)$  is the letter  $i$  in word  $x$ . For  $n=4$  9 words are:  $aaaa$ ,  $aaab$ ,  $abaa$ ,  $abab$ ,  $abba$ ,  $baaa$ ,  $baab$ ,  $abba$ ,  $abba$ . - [Alois P. Heinz](#), May 26 2012  
Number of length- $n$  restricted growth strings (RGS)  $(r(1), r(2), \dots, r(n))$  such that  $r(1)=1$ ,  $r(k) \leq k$ , and  $r(k) \leq r(k-1)$ ; for example, the 9 RGS for  $n=4$  are 1010, 1012, 1201, 1210, 1212, 1230, 1232, 1234, 1234. - [Joerg Arndt](#), Apr 20 2013  
Number of length- $n$  restricted growth strings (RGS)  $(r(1), r(2), \dots, r(n))$  such that  $r(1)=0$ ,  $r(k) \leq k$  and  $r(k) \leq r(k-1)$  is 1; for example, the 9 RGS for  $n=4$  are 0000, 0002, 0003, 0004, 0022, 0024, 0032, 0222, 0224. - [Joerg Arndt](#), Apr 17 2013  
Number of  $(4231,5276143)$ -avoiding involutions in  $S_n$ . - [Alexander Burstein](#), Mar 05 2014  
 $a(n)$  is the number of increasing unary-binary trees with  $n$  nodes that have an associated permutation that avoids 132. For more information about unary-binary trees with associated permutations, see [A258888](#).  
[Manda Riehl](#), Aug 07 2014

From [Tony Foster III](#), Jul 28 2016: (Start)  
A series created using  $2a(n) + a(n+1)$  has Hankel transform of  $F(2n)$ , offset 3,  $F$  being the Fibonacci bisection, [A081386](#) (empirical observation).  
A series created using  $2a(n) + 3a(n+1) + a(n+2)$  gives the Hankel transform of  $\text{Sum}(k=0,n) k! \text{Fibonacci}(2k)$ , offset 3, [A197499](#) (empirical observation). (End)  
Conjecture:  $(2/n) \text{Sum}(k=1,n) (2k+1)k!k^2$  is an integer for each positive integer  $n$ . - [Zhi-Wei Sun](#), Nov 16 2012  
The Rubey and Stump reference proves a refinement of a conjecture of René Marczinkin, which they state as: "The number of 2-Gorenstein algebras which are Nakayama algebras with  $n$  simple modules and have an orientation line as associated quiver equals the number of Motzkin paths of length  $n$ ." - [Eric M. Schmidt](#), Dec 16 2017  
Number of  $U$ -( $k$ )-equivalence classes of lukasiewicz paths. lukasiewicz paths are  $P$ -equivalent if the positions of pattern  $p$  are identical in these paths. [Sergey Kirilov](#), Apr 08 2018  
If  $\text{tau}_1$  and  $\text{tau}_2$  are two distinct permutation patterns chosen from the set  $\{(32,231,312)\}$ , then  $a(n)$  is the number of valid hook configurations of permutations of  $\{1,n\}$  that avoid the patterns  $\text{tau}_1$  and  $\text{tau}_2$ . - [Colin Defant](#), Apr 28 2019  
Number of permutations of length  $n$  that are sorted to the identity by a consecutive-321-avoiding stack followed by a classical-21-avoiding stack. - [Colin Defant](#), Aug 29 2020  
From [Helmut Prodinger](#), Dec 13 2020: (Start)  
The number of paths in the first quadrant starting at  $(0,0)$  and consisting of  $n$  steps from the infinite set  $\{(1,1), (1,-1), (1,-2), (1,-3), \dots\}$ .  
For example, denoting  $U(1,1)$ ,  $D(1,-1)$ ,  $D_2(1,-2)$  for  $j=2$ ,  $a(4)$  counts  $UUUU$ ,  $UUUD$ ,  $UUUD_2$ ,  $UUUD_3$ ,  $UUUD$ ,  $UUUD$ ,  $UUUD$ .  
This step set is inspired by  $\{(1,1), (1,-1), (1,-3), (1,-5), \dots\}$ , suggested by Emeric Deutsch around 2000. See Prodinger link that contains a bijection to Motzkin paths. (End)  
Named by Donaghy (1977) after the Israeli-American mathematician Theodore Motzkin (1908-1970). In Sloane's "A Handbook of Integer Sequences" (1973) they were called "generalized ballot numbers". - [Asirfan Fikder](#), Apr 15 2021  
Number of Motzkin  $n$ -paths  $a(n)$  is split into [A017582\(n\), number of even Motzkin  \$n\$ -paths, and \[A434386\\(n\\), number of odd Motzkin  \\$n\\$ -paths. The value \\[A017582\\\(n\\\) - A434386\\\(n\\\) can be called the "shadow" of  \\\$a\\\(n\\\)\\\$  \\\(see \\\[A434373\\\]\\\(#\\\)\\\). - \\\[Gennady Eremin\\\]\\\(#\\\), May 17 2021  
Conjecture: if  \\\$p\\\$  is a prime of the form  \\\$6m+1\\\$  \\\(\\\[A002476\\\\), then  \\\\$a\\\\(p-2\\\\)\\\\$  is divisible by  \\\\$p\\\\$ . Currently, no counterexample exists for  \\\\$p < 10^7\\\\$ . Personal communication from \\\\[Robert Gerbicz\\\\]\\\\(#\\\\): mod such  \\\\$p\\\\$  this is equal to  \\\\$a\\\\(6p-7\\\\) \pmod{p}\\\\$  with comment: "Every \\\\[A066796\\\\\(n\\\\\) from \\\\\[A066796\\\\\\(p-1\\\\\\)/2 is divisible by prime  \\\\\\$p\\\\\\$  of form  \\\\\\$6m+1\\\\\\$ ". - \\\\\\[Serge Bateman\\\\\\]\\\\\\(#\\\\\\), Feb 08 2022  
From \\\\\\[Rob Burns\\\\\\]\\\\\\(#\\\\\\), Nov 11 2024: \\\\\\(Start\\\\\\)  
The number of paths is proved to be equal to the number of paths by Rob Burns in the Links below. The result is contained in Tables 4 and 5 of the paper, which show that  \\\\\\$a\\\\\\(p-2\\\\\\) \equiv 0 \pmod{p}\\\\\\$  when  \\\\\\$p \equiv 1 \pmod{6}\\\\\\$  and  \\\\\\$a\\\\\\(p-2\\\\\\) \equiv -1 \pmod{p}\\\\\\$  when  \\\\\\$p \equiv -1 \pmod{6}\\\\\\$ .  
In fact, the 2012 paper by Burns establishes several general congruences for  \\\\\\$a\\\\\\(p^k - 2\\\\\\)\\\\\\$  where  \\\\\\$k \geq 1\\\\\\$ .  
If  \\\\\\$p \equiv 1 \pmod{6}\\\\\\$  then  \\\\\\$a\\\\\\(p^k - 2\\\\\\) \equiv 0 \pmod{p}\\\\\\$  for  \\\\\\$k \geq 1\\\\\\$ .  
If  \\\\\\$p \equiv -1 \pmod{6}\\\\\\$  then  \\\\\\$a\\\\\\(p^k - 2\\\\\\) \equiv -1 \pmod{p}\\\\\\$  when  \\\\\\$k\\\\\\$  is odd and  \\\\\\$a\\\\\\(p^k - 2\\\\\\) \equiv 0 \pmod{p}\\\\\\$  when  \\\\\\$k\\\\\\$  is even. These are consequences of the transitions provided in Tables 4, 5 and 6 of the paper.  
The 2024 paper by Nadav Kohen also proves the conjecture. Proposition 5 of the paper states that a prime  \\\\\\$p\\\\\\$  divides  \\\\\\$a\\\\\\(p-2\\\\\\)\\\\\\$  if and only if  \\\\\\$p \equiv 1 \pmod{3}\\\\\\$ . \\\\\\(End\\\\\\)  
From \\\\\\[Peter Bal\\\\\\]\\\\\\(#\\\\\\), Feb 10 2022: \\\\\\(Start\\\\\\)  
Conjectures:  
\\\\\\(1\\\\\\) For prime  \\\\\\$p \equiv 1 \pmod{6}\\\\\\$  and  \\\\\\$r, r \equiv 1\\\\\\$ ,  \\\\\\$a\\\\\\(p^{r-2}\\\\\\) - 2 = -A085717\\\\\\(n-1\\\\\\) \pmod{p}\\\\\\$ , where take \\\\\\[A085717\\\\\\\(0\\\\\\\) = 0 to match Batolov's conjecture above.  
\\\\\\\(2\\\\\\\) For prime  \\\\\\\$p \equiv 5 \pmod{6}\\\\\\\$  and  \\\\\\\$r \equiv 1\\\\\\\$ ,  \\\\\\\$a\\\\\\\(p^{r-2}\\\\\\\) = -A085717\\\\\\\(n\\\\\\\) \pmod{p}\\\\\\\$ .  
\\\\\\\(3\\\\\\\) For prime  \\\\\\\$p \equiv 3 \pmod{6}\\\\\\\$  and  \\\\\\\$r \equiv 1\\\\\\\$ ,  \\\\\\\$a\\\\\\\(p^{r-2}\\\\\\\) = a\\\\\\\(n\\\\\\\) \pmod{p}\\\\\\\$  for  \\\\\\\$0 \leq n \leq \\\\\\\(p^k - 3\\\\\\\)\\\\\\\$ .  
\\\\\\\(4\\\\\\\) For prime  \\\\\\\$p \equiv 5\\\\\\\$  and  \\\\\\\$k \geq 2\\\\\\\$ ,  \\\\\\\$a\\\\\\\(n + p^k\\\\\\\) \equiv a\\\\\\\(n\\\\\\\) \pmod{p^2}\\\\\\\$  for  \\\\\\\$0 \leq n \leq \\\\\\\(p^{k-1} - 3\\\\\\\)\\\\\\\$ . \\\\\\\(End\\\\\\\)  
The Hankel transform of this sequence with  \\\\\\\$a\\\\\\\(0\\\\\\\)\\\\\\\$  omitted gives the period-6 sequence  \\\\\\\$1, 0, -1, -1, 0, 1\\\\\\\$ . The number of terms in which the exponent of any variable  \\\\\\\$x\\\\\\\_i\\\\\\\$  is not greater than 2 in the expansion of  \\\\\\\$\text{Product}\\\\\\\(j=1,n\\\\\\\) \text{Sum}\\\\\\\(i=1,j\\\\\\\) x\\\\\\\_i\\\\\\\$ . E.g.:  \\\\\\\$a\\\\\\\(4\\\\\\\) = 9\\\\\\\$ :  \\\\\\\$3x\\\\\\\_1^2x\\\\\\\_2x\\\\\\\_3^2 + 4x\\\\\\\_1x\\\\\\\_2^2x\\\\\\\_3x\\\\\\\_4 + 2x\\\\\\\_1x\\\\\\\_2^2x\\\\\\\_3x\\\\\\\_4 + x\\\\\\\_1^2x\\\\\\\_3^2x\\\\\\\_4^2 + 2x\\\\\\\_1x\\\\\\\_3^2x\\\\\\\_4^2 + x\\\\\\\_1x\\\\\\\_4^2x\\\\\\\_3^2 + x\\\\\\\_1x\\\\\\\_4^2x\\\\\\\_3^2 + x\\\\\\\_1x\\\\\\\_4^2x\\\\\\\_3^2 + x\\\\\\\_1x\\\\\\\_4^2x\\\\\\\_3^2\\\\\\\$ . - \\\\\\\[Ellis Raser\\\\\\\]\\\\\\\(#\\\\\\\), Dec 20 2024  
The Hankel matrix corresponding to this sequence factorizes as  \\\\\\\$T \cdot \text{Transpose}\\\\\\\(T\\\\\\\)\\\\\\\$ , where  \\\\\\\$T\\\\\\\$  is the lower unitriangular array \\\\\\\[A065189\\\\\\\]\\\\\\\(#\\\\\\\) \\\\\\\(row reverse of the Motzkin triangle \\\\\\\[A026380\\\\\\\]\\\\\\\(#\\\\\\\)\\\\\\\). Hence the Hankel determinants are all equal to 1 as stated above by Michael Somos. - \\\\\\\[Peter Bal\\\\\\\]\\\\\\\(#\\\\\\\), Nov 23 2025  
From \\\\\\\[Michael Jung\\\\\\\]\\\\\\\(#\\\\\\\), Jul 29 2026: \\\\\\\(Start\\\\\\\)  
Number of 2-sorted-sortable permutations of  \\\\\\\$\{1,2,\dots,n+1\}\\\\\\\$  ending with 1.  
Number of standard composition tableaux of width at most 2 on  \\\\\\\$n\\\\\\\$  cells. \\\\\\\(End\\\\\\\)\\\\\\]\\\\\\(#\\\\\\)\\\\\]\\\\\(#\\\\\)\\\\]\\\\(#\\\\)\\\]\\\(#\\\)\\]\\(#\\)\]\(#\)](#)

## Theorem (L. 2026)

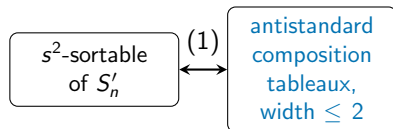
*# of  $s^2$ -sortable permutations in  $S'_n$  is  $n$ th Motzkin number.*

Idea: matching!

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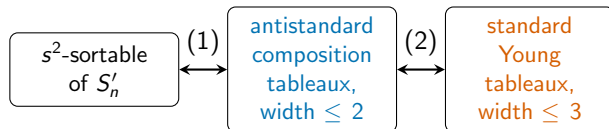
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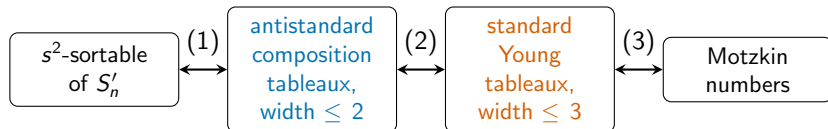


# Proof idea

## Theorem (L. 2026)

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## Lemma (Chen–L.–Zhang 2025, Zhang 2026)

Take  $\pi \in S'_n$  and apply  $s$  repeatedly. Then

- each time, the *right-to-left maxima* of the part before 0 go after 0 and
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$$4\mathbf{6}521\mathbf{3}0 \xrightarrow{s} 412\mathbf{0}356$$

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$$4120\mathbf{3}56 \xrightarrow{s} 10\mathbf{2}3456$$

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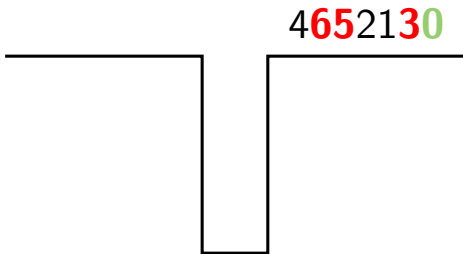
$$\mathbf{1}023456 \xrightarrow{s} 0\mathbf{1}23456$$

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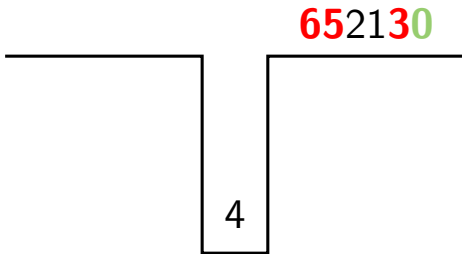


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$$4\mathbf{65}21\mathbf{30} \xrightarrow{s} 412\mathbf{0356}$$

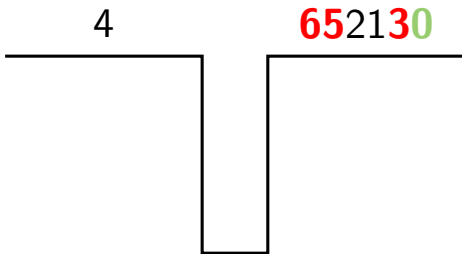


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$$4652130 \xrightarrow{s} 4120356$$

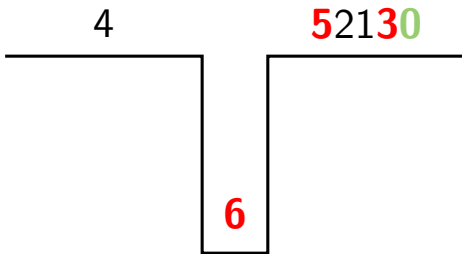


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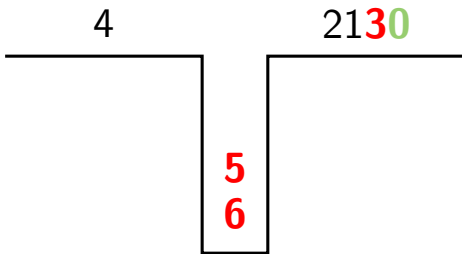


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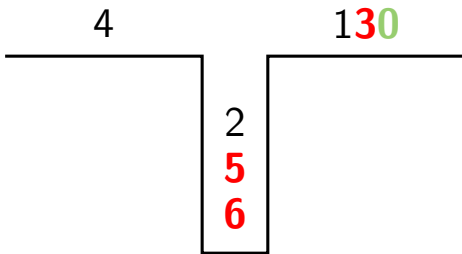


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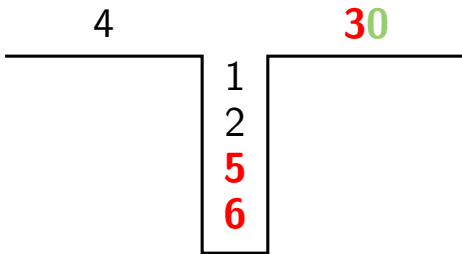


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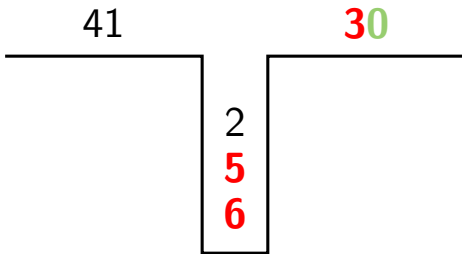


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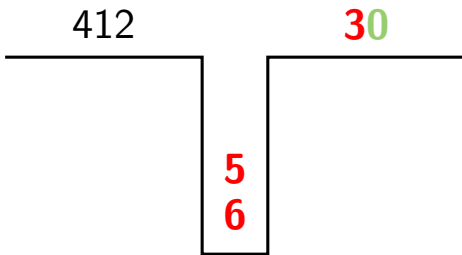


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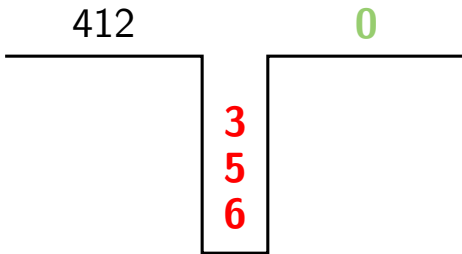


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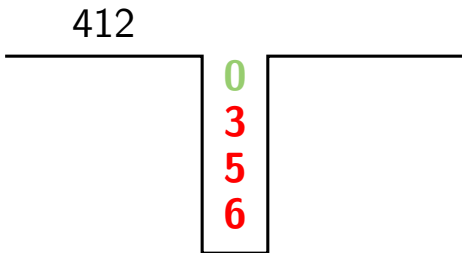


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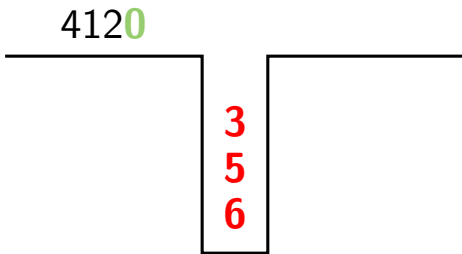


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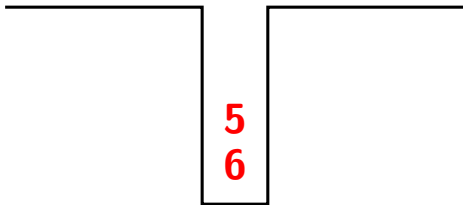
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41203



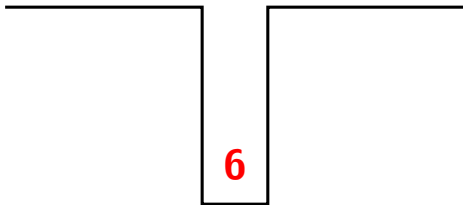
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$$4\mathbf{6}521\mathbf{3}0 \xrightarrow{s} 412\mathbf{0}35\mathbf{6}$$

412035



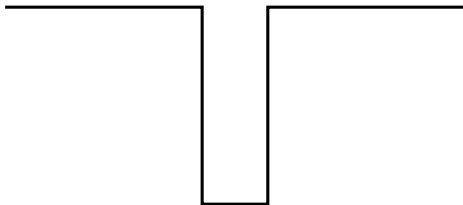
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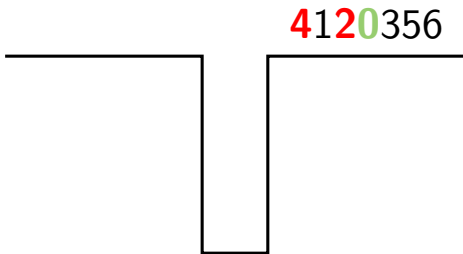


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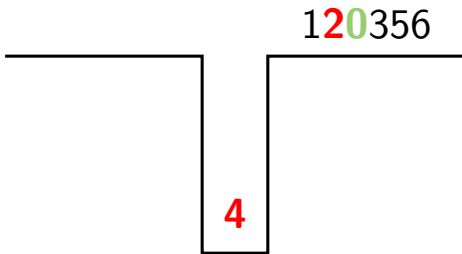


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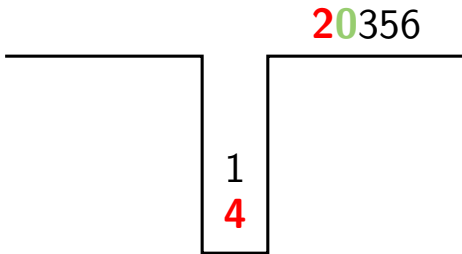


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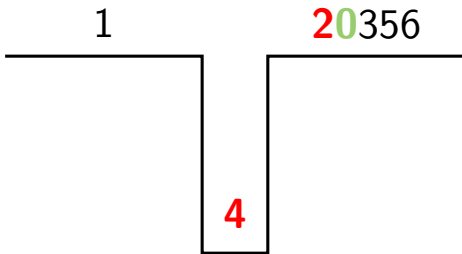


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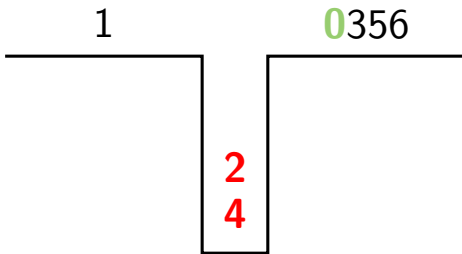


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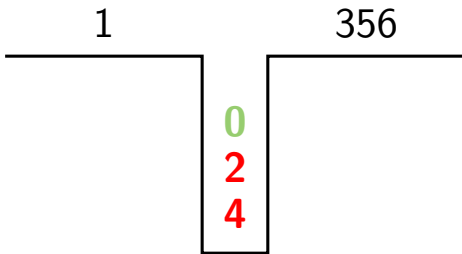


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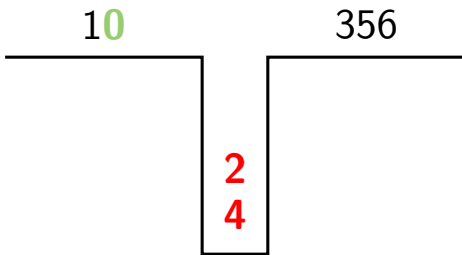


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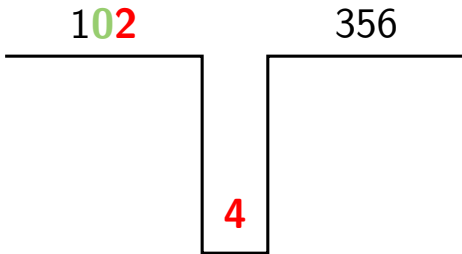


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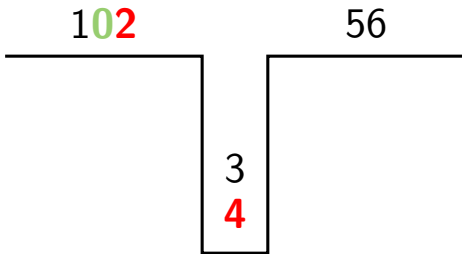


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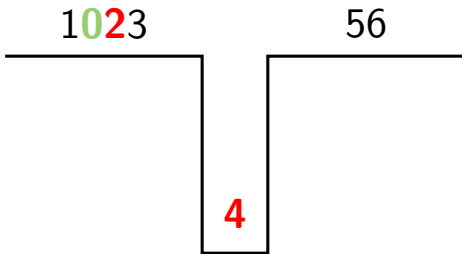


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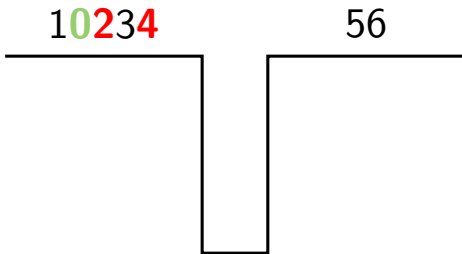


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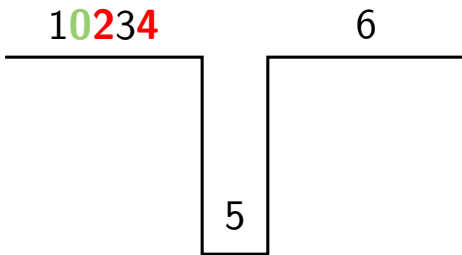


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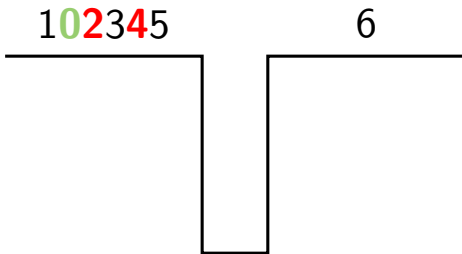


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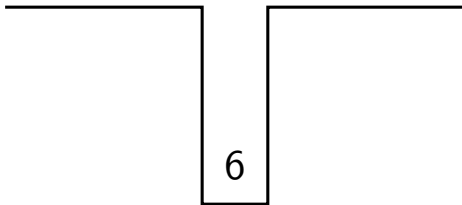
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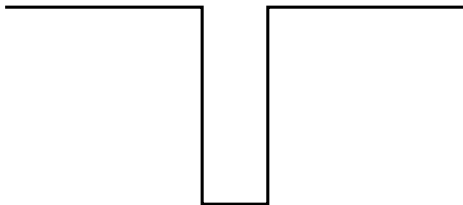
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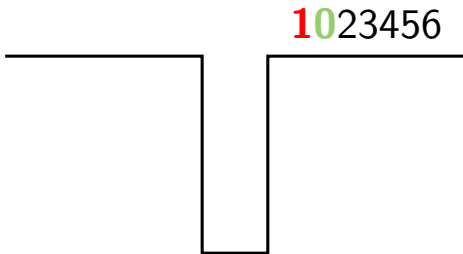


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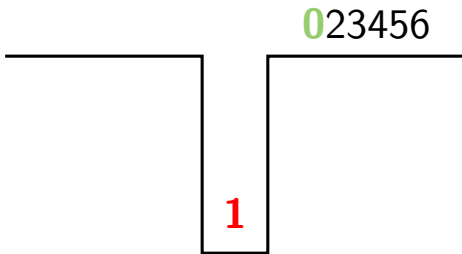


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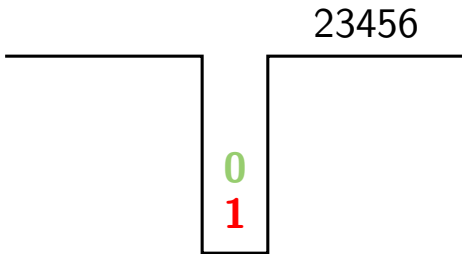


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Take  $\pi \in S'_n$  and apply  $s$  repeatedly. Then

- each time, the *right-to-left maxima* of the part before 0 go after 0 and
- the part after 0 is always sorted.

$$1023456 \xrightarrow{s} 0123456$$

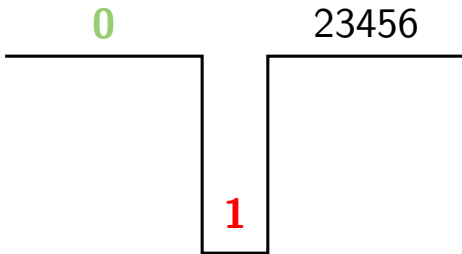


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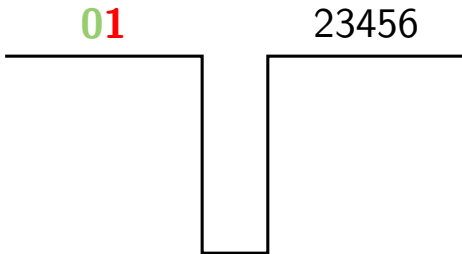


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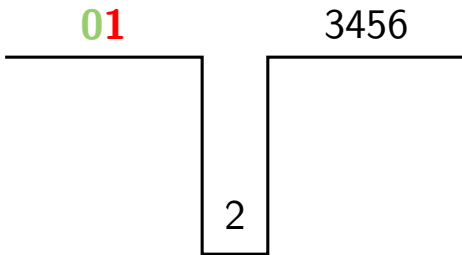


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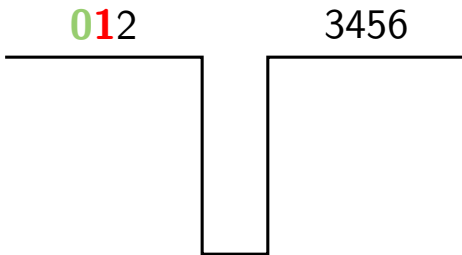


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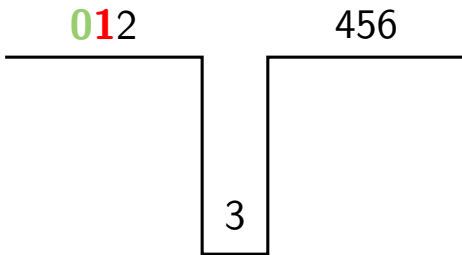


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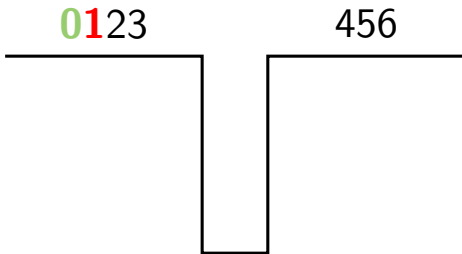


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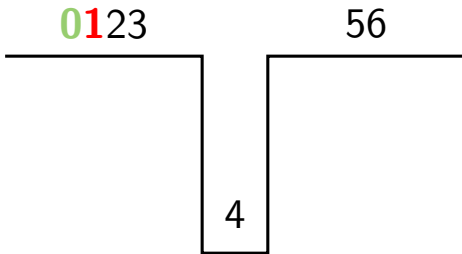


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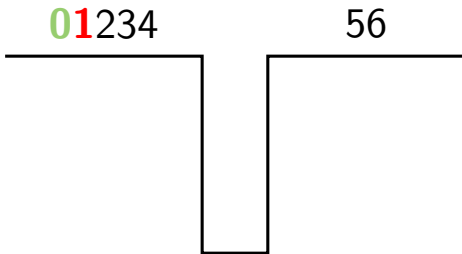


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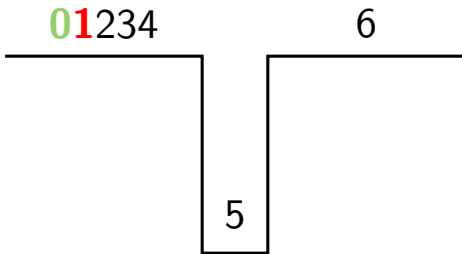


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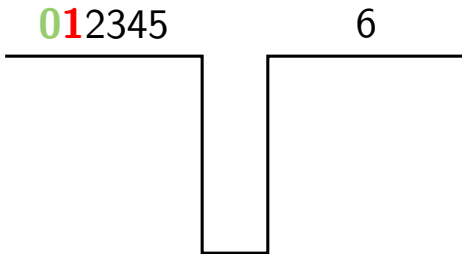


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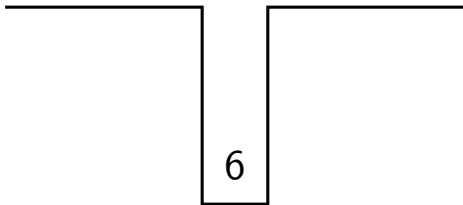
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012345



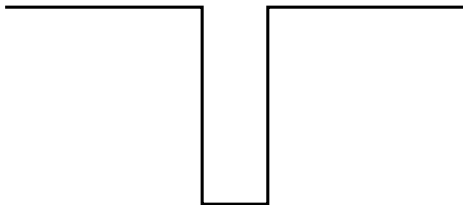
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0123456



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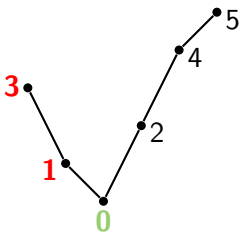
- each time, the *right-to-left maxima* of the part before 0 go after 0 and
- the part after 0 is always sorted.

$$4\mathbf{6}521\mathbf{3}0 \xrightarrow{s} 4120\mathbf{3}56$$

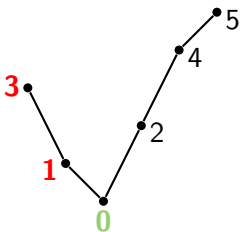
$$\mathbf{4}120\mathbf{3}56 \xrightarrow{s} 10\mathbf{2}3\mathbf{4}56$$

$$\mathbf{1}023456 \xrightarrow{s} 0\mathbf{1}23456$$

**31****0**245 is s-sortable since all elements before 0 are right-to-left maxima

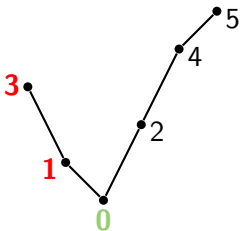


**31**0245 is  $s$ -sortable since all elements before 0 are right-to-left maxima

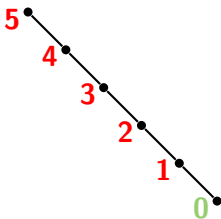


$s$ -sortable permutations are V-shaped!

**31**0245 is  $s$ -sortable since all elements before 0 are right-to-left maxima



$s$ -sortable permutations are V-shaped! Only one  $s$ -sortable in  $S'_n$ :



If  $\pi \in S'_n$  is  $s^2$ -sortable,  $\leq 1$  elements between RL-maxima of  $\pi$

42310

good

42130

bad

If  $\pi \in S'_n$  is  $s^2$ -sortable,  $\leq 1$  elements between RL-maxima of  $\pi$

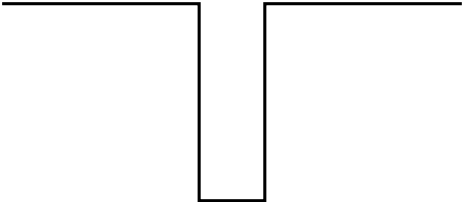
42310

good

42130

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42130



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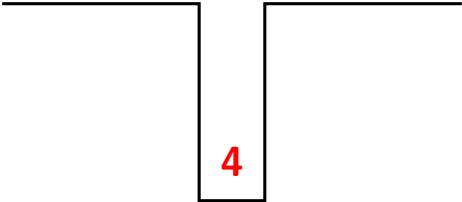
42310

good

42130

bad

2130



4

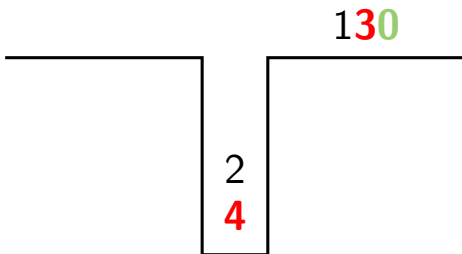
If  $\pi \in S'_n$  is  $s^2$ -sortable,  $\leq 1$  elements between RL-maxima of  $\pi$

42310

good

42130

bad



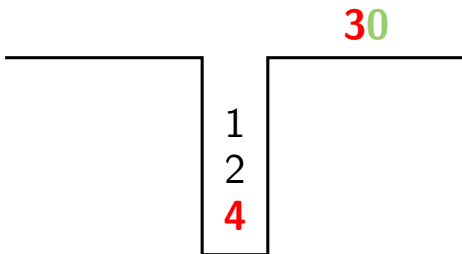
If  $\pi \in S'_n$  is  $s^2$ -sortable,  $\leq 1$  elements between RL-maxima of  $\pi$

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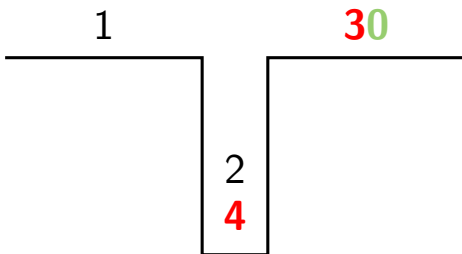
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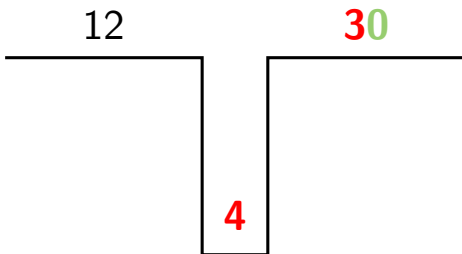
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42310

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42130

bad



If  $\pi \in S'_n$  is  $s^2$ -sortable,  $\leq 1$  elements between RL-maxima of  $\pi$

42310

good

42130

bad

12

0

3

4

If  $\pi \in S'_n$  is  $s^2$ -sortable,  $\leq 1$  elements between RL-maxima of  $\pi$

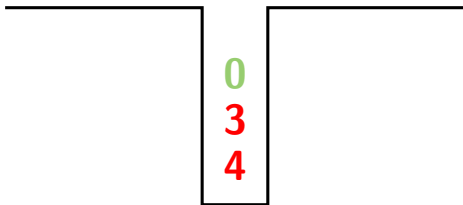
42310

good

42130

bad

12



If  $\pi \in S'_n$  is  $s^2$ -sortable,  $\leq 1$  elements between RL-maxima of  $\pi$

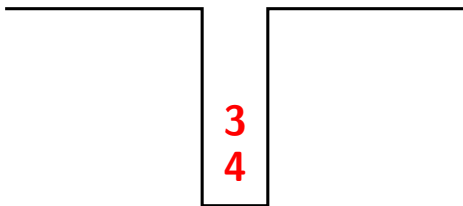
42310

good

42130

bad

120



If  $\pi \in S'_n$  is  $s^2$ -sortable,  $\leq 1$  elements between RL-maxima of  $\pi$

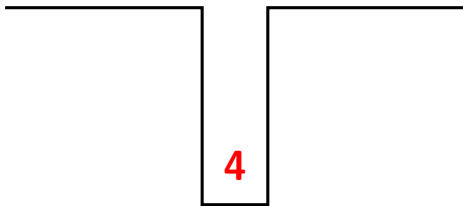
42310

good

42130

bad

1203



If  $\pi \in S'_n$  is  $s^2$ -sortable,  $\leq 1$  elements between RL-maxima of  $\pi$

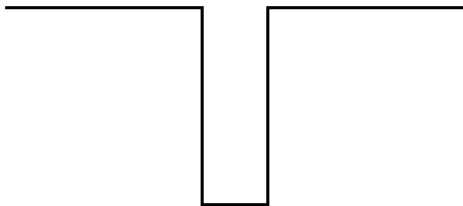
42310

good

42130

bad

12034



354120

487635120

3**5**41**2**0



|   |   |
|---|---|
| 5 | 3 |
| 4 |   |
| 2 | 1 |

4**8**7**6**3**5**1**2**0



|   |   |
|---|---|
| 8 | 4 |
| 7 |   |
| 6 |   |
| 5 | 3 |
| 2 | 1 |

354120



|   |   |
|---|---|
| 5 | 3 |
| 4 |   |
| 2 | 1 |

487635120



|   |   |
|---|---|
| 8 | 4 |
| 7 |   |
| 6 |   |
| 5 | 3 |
| 2 | 1 |

- Rows and columns decreasing
- Entries are exactly  $\{1, 2, \dots, n\}$
- Width at most 2

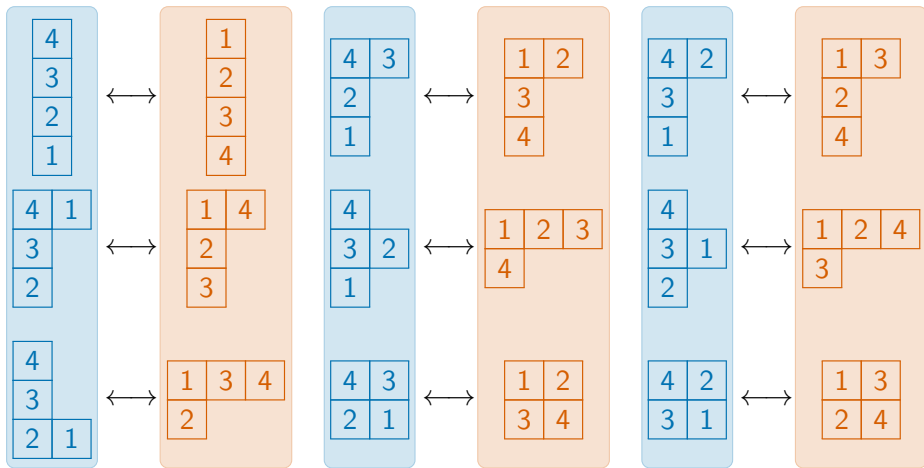
“Antistandard composition tableaux with width at most 2”

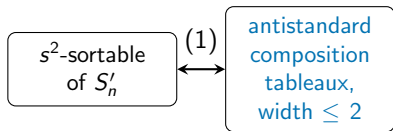


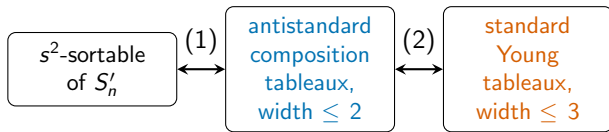
My result:

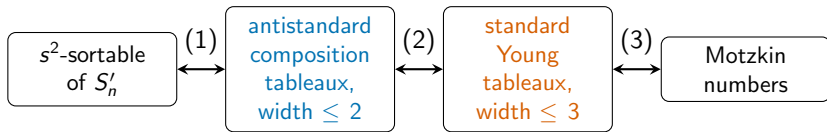
antistandard composition tableaux, width  $\leq 2 \longleftrightarrow$

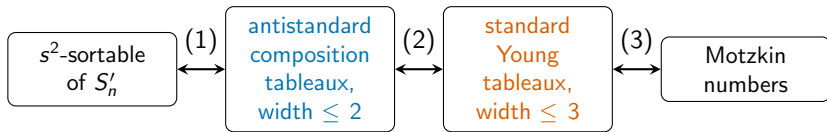
standard Young tableaux, width  $\leq 3$











## Theorem (L. 2026)

*# of  $s^2$ -sortable permutations in  $S'_n$  is  $n$ th Motzkin number.*

*n*th Motzkin counts:

Number of 4321-, {3412, 2413}, {3412, 3142}- and 3412-avoiding involutions in  $S_n$ .  
 Number of sequences of length  $n-1$  consisting of positive integers such that the first and last elements are 1 or 2 and the absolute difference between any 2 consecutive elements is 0 or 1. - [Jon Perry](#), Sep 04 2003  
 From [David Callan](#), Jul 15 2004: (Start)  
 Also number of Motzkin  $n$ -paths: paths from  $(0,0)$  to  $(n,0)$  in an  $n \times n$  grid using only steps  $U = (1,1)$ ,  $F = (1,0)$  and  $D = (1,-1)$ .

Number of Dyck  $n$ -paths with no  $UUU$ . (Given such a Dyck  $n$ -path, change each  $UUD$  to  $U$ , then change each remaining  $UD$  to  $F$ . This is a bijection to Motzkin  $n$ -paths. Example with  $n=5$ :  $UUDUDUUDDD \rightarrow UFUUDDD$ .)

[illegible]

$a(n)$  = number of Dyck  $n$ -paths all of whose valleys have even  $x$ -coordinate (when path starts at origin). For example,  $T(4,2)=3$  counts UDUUDD, UDUUDD, UDUUDD. Given such a path, split it into  $n$  subpaths of length 2 and transform UU $\rightarrow$ U, DD $\rightarrow$ D, UD $\rightarrow$ F (there will be no DFs for that would entail a valley with odd  $x$ -coordinate). This is a bijection to Motzkin  $n$ -paths. — David Callan, Jun 07 2006

Also the number of standard Young tableaux of height  $\leq 3$ . - Mike Zabrocki, Mar 24 2007

a(n) is the number of RNA shapes of size  $2n+2$ . RNA Shapes are essentially Dyck words without "directly nested" motifs of the form  $A[B]C$ , for  $A, B$  and  $C$  Dyck words. The first RNA Shapes are  ${}_{[1]}[1]_{[1]}$ ,  ${}_{[1]}[1][1]_{[1]}$ ,  ${}_{[1]}[1][1][1]_{[1]}$ ,  ${}_{[1]}[1][1][1][1]_{[1]}$ ,  ${}_{[1]}[1][1][1][1][1]_{[1]}$ , ... - Yann Ponty (AT) [r1, f1], May 30 2007

Yann Ponty is a bioinformatician, and he generates RNA shapes by going to  $(1,1)$  and  $(-1,1)$  the same sequence but starting in  $(0,0)$  and going to the right. Take dot product of  $A$  and  $B$  and add the result to  $n$ -th term of  $A$  to get the  $(n+1)$ -th term of  $A$ . Example:  $a(5) = 21$  as follows: Take dot product of  $A = (0, 0, 4, 2, 1)$

1, 1) and  $\{0, 1, 1, 2, 4\} = \{0, +4 + 2 + 2 + 4\} = 12$ ; which is added to  $9 = 21$ . - Gary W. Adamson, Oct 27 2008

Dec 21 2008

Starting with offset 1 iterates of  $M = [1, 0, 0, 9, \dots]$ , where  $M$  is a tridiagonal matrix with  $\{0, 1, 1, \dots\}$  in the main diagonal and  $\{1, 1, 1, \dots\}$  in the super and subdiagonals. Gary W. Adamson, Jan 07 2009

If  $p'$  is a permutation of  $\{1, 2, \dots, n\}$  with genus  $g$ , then  $(p')^{-1} p' = (1, 2, \dots, n)$ . If  $p'$  is defined by  $p'_i = (i-2)! [n-i+1] z(p')$ , where  $p'$  is the inverse permutation of  $p$ ,  $c = 2341\dots n = (1, 2, \dots, n)$ , and  $z(q)$  is the number of cycles of the permutation  $q$ . Example:  $a(4) = 9$ ; indeed,  $p = (1, 2, 3, 4)$  has genus 1, so  $p'_1 = 0!$  and  $p'_2 = 1!$ . The permutation  $p' = (1, 2, 3, 4)$  has genus 0 if and only if the cycle decomposition of  $p$  gives a noncrossing partition of  $\{1, 2, \dots, n\}$  and each cycle of  $p$  is increasing (see Lemma 2.1 of the Duluc-Simion reference). (Also, redundantly, for  $n=3412$  we have  $p_c = 2341=3412=1432$  and  $p'_c = (1/2)(4! - 4 - 2 - 1) = 6$ .)

End.

Let  $w(i, j, n)$  denote walks in  $\mathbb{N}^2$  which satisfy the multivariate recurrence  $w(i, j, n) = w(i, j+1, n-1) + w(i-1, j, n-1) + w(i+1, j-1, n-1)$  with boundary conditions  $w(0, 0, 0) = 1$  and  $w(i, j, n) = 0$  if  $i$  or  $j$  or  $n$  is  $< 0$ . Then  $a(n) = \sum_{i=0..n} \sum_{j=0..n} w(i, j, n)$  is the number of such walks of length  $n$ . -

(n)  $\lfloor n-1 \rfloor$  tends to  $3.0$  as  $N \rightarrow \infty$ :  $1+2\cos(2\pi/N)$  relating to longest odd  $N$  regular polygon diagonals, by way of example,  $N=7$ : Using the triangular generator [cf. comment in Jan 07 2009], for polygon  $N=7$ , we extract an  $(N-1)/2 = 3 \times 3$  matrix,  $\{0,1,0\}$ ,  $\{0,1,1\}$ ,  $\{0,1,1\}$  with an e-val of  $2.46967\dots$ ; the longest Heptagon diagonal with length  $= 3$ . As  $N$  tends to infinity, the diagonal lengths tend to  $3.0$ , the cosine of  $\pi/3$  of the sequence. — [Gary W. Adamson](#), June 2011

Number of  $\lfloor n-1 \rfloor$ -length permutations avoiding the pattern  $132$ :  $1$  and the dotted pattern  $231$ dot $\{1\}$ . — [Jean-Luc Baril](#), Mar 2012

Number of  $n$ -length words over alphabet  $\{a,b,c\}$  such that for every prefix  $z$  of  $w$  we have  $\#(z,a) = \#(z,b) = \#(z,c)$ , where  $\#(z,x)$  counts the letters  $x$  in word  $z$ . The  $a(4) = 9$  words are:  $aaaa, aaba, abaa, abba, abab, abab, abca, abca$ . - **Alois P. Heinz**, May 26 2012

Number of length- $n$  restricted growth strings (RGS)  $[r(1), r(2), \dots, r(n)]$  such that  $r(1)=1, r(k) \leq k$ , and  $r(k) = r(k-1)$ ; for example, the 9 RGS for  $n=4$  are 1010, 1012, 1201, 1210, 1212, 1230, 1231, 1232, 1234. - **Joerg Arndt**, Aug 16 2003.

Number of length- $n$  restricted growth strings (RGS)  $[r(1), r(2), \dots, r(n)]$  such that  $r(1)=0$ ,  $r(k) \leq k$  and  $r(k)-r(k-1) \in \{0, 1\}$ ; for example, the 9 RGS for  $n=4$  are 0000, 0002, 0003, 0004, 0022, 0024, 0033, 0222, 0224.

a(n) is the number of increasing unary-binary trees with  $n$  nodes that have an associated permutation that avoids 132. For more information about unary-binary trees with associated permutations, see [A245888](#). - [Manda Riehl](#), Aug 07 2014

From [Tony Foster III](#), Jul 28 2016: (Start)  
 A series created using  $2a(n) + a(n+1)$  has Hankel transform of  $F(2n)$ , offset 3,  $F$  being the Fibonacci bisection, [A081986](#) (empirical observation).  
 A series created using  $2a(n) + 3a(n+1) + a(n+2)$  gives the Hankel transform of  $\text{Sum}(k=0..n) k \cdot \text{Fibonacci}(2k)$ , offset 3, [A197659](#) (empirical observation). (End)  
 Conjecture:  $(2/n) \cdot \text{Sum}(k=1..n) (2k+1) \cdot a(k)^2$  is an integer for each positive integer  $n$ . - [Zhi-Wei Sun](#), Nov

The Rubey and Stump reference proves a refinement of a conjecture of René Marczinzik, which they state as:  
 "The number of 2-Gorenstein algebras which are Nakayama algebras with  $n$  simple modules and have an oriented line as associated quiver equals the number of Motzkin paths of length  $n$ ." - [Eric M. Schmidt](#), Dec 16 2017

Number of  $U_{\{k\}}$ -equivalence classes of Łukasiewicz paths. Łukasiewicz paths are  $P$ -equivalent iff the positions of pattern  $P$  are identical in these paths. - [Sergey Kirgizov](#), Apr 08 2018

If  $\tau_1$  and  $\tau_2$  are two distinct permutation patterns chosen from the set  $\{132, 231, 312\}$ , then  $a(n)$  is the number of valid hook configurations of permutations of  $[n+1]$  that avoid the patterns  $\tau_1$  and  $\tau_2$ . - [Colin Defant](#), Apr 28 2019

Number of permutations of length  $n$  that are sorted to the identity by a consecutive-321-avoiding stack followed by a classical-21-avoiding stack. - [Colin Defant](#), Aug 29 2020

a(n) is the number of paths in the first quadrant starting at (0,0) and consisting of n steps from the infinite set  $\{(1,1), (1,-1), (1,-2), (1,-3), \dots\}$ .

For example, denoting  $U=(1,1)$ ,  $D=(1,-1)$ ,  $D_{-j}=(1,-j)$  for  $j \geq 2$ ,  $a(4)$  counts  $UUUU$ ,  $UUUD$ ,  $UUUD_2$ ,  $UUUD_3$ ,  $UUUD$ ,  $UUDD$ ,  $UUD_2U$ ,  $UUDU$ ,  $UUDU$ .

This step set is inspired by  $\{(1,1), (1,-1), (1,-3), (1,-5), \dots\}$ , suggested by Émeric Deutsch around 2000. See Prodinger link that contains a bijection to Motzkin paths. (End)

Named by Donaghey (1977) after the Israeli-American mathematician Theodore Motzkin (1908–1970). In Sloane's "A Handbook of Integer Sequences" (1973) they were called "generalized ballot numbers". - [Amiram Eldar](#), Apr 15 2021

Number of Motzkin  $n$ -paths  $a(n)$  is split into [A107587](#)( $n$ ), number of even Motzkin  $n$ -paths, and [A343386](#)( $n$ ), number of odd Motzkin  $n$ -paths. The value [A107587](#)( $n$ ) - [A343386](#)( $n$ ) can be called the "shadow" of  $a(n)$  (see [A343773](#)). - Gennady Fremin, May 17 2021

Conjecture: If  $p$  is a prime of the form  $6m+1$  ([A002476](#)), then  $a(p-2)$  is divisible by  $p$ . Currently, no counterexample exists for  $p < 10^7$ . Personal communication from [Robert Gerbicz](#): mod such  $p$  this is  $\text{count}(n \leq p, a(n) \equiv 0 \pmod{p}) = \frac{p-1}{6}$ . [A066706](#) is  $\sum_{n \leq x} a(n) \pmod{p}$ . [A066706](#) is  $\frac{p-1}{6}$  for  $p \equiv 1 \pmod{6}$ .

From [Rob Burns](#), Nov 11 2024: (Start)

The conjecture is proved in the 2017 paper by Rob Burns in the links below. The result is contained in Tables 4 and 5 of the paper, which show that  $a(p-2) \equiv 0 \pmod{p}$  when  $p \equiv 1 \pmod{6}$  and  $a(p-2) \equiv -1 \pmod{p}$  when  $p \equiv -1 \pmod{6}$ .

In fact, the 2017 paper by Burns establishes more general congruences for  $a(p^k - 2)$  where  $k \geq 1$ . If  $p \equiv 1 \pmod{6}$  then  $a(p^k - 2) \equiv 0 \pmod{p}$  for  $k \geq 1$ .

If  $p \equiv -1 \pmod 6$  then  $a(p^k - 2) \equiv -1 \pmod p$  when  $k$  is odd and  $a(p^k - 2) \equiv 0 \pmod p$  when  $k$  is even. These are consequences of the transitions provided in Tables 4, 5 and 6 of the paper. The 2024 paper by Nadia Kehan also proves the conjecture. Proposition 6 of the paper states that a prime  $p$

The 2024 paper by Madya Kohen also proves the conjecture. Proposition 6 of the paper states that a prime  $p$  divides  $a(p-2)$  if and only if  $p \equiv 1 \pmod{3}$ . (End)  
From [Peter Bala](#), Feb 10 2022: (Start)

Conjectures:  
(1) For prime  $p \equiv 1 \pmod{6}$  and  $n, r \geq 1$ ,  $a(n+p^r-2) \equiv -A005717(n-1) \pmod{p}$ , where we take  $A005717(0) = 0$  to match Batalov's conjecture above.

```
(2) For prime  $p \equiv 5 \pmod{6}$  and  $n \geq 1$ ,  $a(n \cdot p - 2) \equiv -A085773(n) \pmod{p}$ .
(3) For prime  $p \geq 3$  and  $k \geq 1$ ,  $a(n + p^k) \equiv a(n) \pmod{p}$  for  $0 \leq n < (p^k - 3)$ .
(4) For prime  $n \equiv 5 \pmod{6}$  and  $k \geq 2$ ,  $a(n + p^k) \equiv a(n) \pmod{p^2}$  for  $k \leq n \leq (p^k(k-1) - 3)$ . (End)
```

The Hankel transform of this sequence with  $a(0)$  omitted gives the period-6 sequence  $[1, 0, -1, -1, 0, 1, \dots]$  which is [A018892](#) with its first term omitted, while the Hankel transform of the current sequence is the all-ones sequence [A000033](#), and also it is the unique sequence with this property which is smaller to

The all-ones sequence [00001111](#), and also it is the unique sequence with this property which is similar to the unique Hankel transform property of the Catalan numbers. - [Michael Somos](#), Apr 17 2022

Product\_{j=1..n} Sum\_{i=1..n} x\_{j,i}. E.g.:  $a(4) = 9: 3x_{11}^2x_{22}x_{32}^2, 4x_{11}^2x_{22}x_{32}x_{33}, 2x_{11}^2x_{22}x_{32}x_{34}, x_{11}^2x_{33}^2x_{34}, x_{11}^2x_{33}x_{34}^2, 2x_{11}x_{12}^2x_{23}x_{33}, x_{11}x_{12}^2x_{23}x_{34}, x_{11}x_{23}^2x_{33}^2, x_{11}x_{23}^2x_{33}x_{34}.$  – [Elif Baser](#), Dec 20 2024

The Hankel matrix corresponding to this sequence factorizes as  $T \times \text{Transpose}(T)$ , where  $T$  is the lower

unitriangular array [A054189](#) (row reverse of the Motzkin triangle [A026308](#)). Hence the Hankel determinants are all equal to 1 as stated above by Michael Somos. - [Peter Bala](#), Nov 23 2025

Number of 2-stack-sortable permutations of  $\{1, 2, \dots, n+1\}$  ending with 1.  
 Number of standard composition tableaux of width at most 2 on  $n$  cells. (End)

*n*th Motzkin counts:

Number of 4321-, {3412, 2413}, {3412, 3142}- and 3412-avoiding involutions in  $S_n$ .  
 Number of sequences of length  $n-1$  consisting of positive integers such that the first and last elements are 1 or 2 and the absolute difference between any 2 consecutive elements is 0 or 1. - [Jon Perry](#), Sep 04 2003  
 From [David Callan](#), Jul 15 2004: (Start)  
 Also number of Motzkin  $n$ -paths: paths from  $(0,0)$  to  $(n,0)$  in an  $n \times n$  grid using only steps  $U = (1,1)$ ,  $F = (1,0)$  and  $D = (1,-1)$ .

Number of Dyck  $n$ -paths with no  $UUU$ . (Given such a Dyck  $n$ -path, change each  $UUD$  to  $U$ , then change each remaining  $UD$  to  $F$ . This is a bijection to Motzkin  $n$ -paths. Example with  $n=5$ :  $UUDUDUUDDD \rightarrow UFUUDDD$ .)

Number of Dyck  $[n]$ -pathsthz no UDU. (Given such a Dyck  $[n]$ -path, mark each  $U$  that is followed by a  $D$  and each  $D$  that is not followed by a  $U$ . Then change each unmarked  $U$  whose matching  $D$  is marked to an  $F$ . Lastly, delete all the marked steps. This is a bijection to Motzkin  $n$ -paths. Example with  $n=6$  and marked steps in **bold** type:  $UDUDUD \rightarrow DUDUDUD \rightarrow DUDUDUD \rightarrow DUDUDUD \rightarrow DUDUDUD \rightarrow DUDUDUD \rightarrow DUDUDUD \rightarrow DUDUDUD$ . (End)

(a)  $n!$  number of strings of length  $2n$  with the following recursively defined set:  $\{ \text{containing exactly one } U \text{ and } n \text{ } D\text{'s, for any strings } a \text{ and } b \text{ in } L \}$ . We also find  $(ab)$  in  $L$ . The first few elements of  $L$  are  $\{(), (U), (UU), (UU), (UUUU), (UUUU), (UUUU), (UUUU), (UUUU)\}$  and so on. This proves that  $a(n)$  is less than or equal to  $A(n)$ , the  $n$ -th Catalan number. See [Orlicki Link \(2024\)](#). – Saul Schleiser

[\[sausch@math.rutgers.edu\]](#), Feb 23 2006 (Additional linked content added by [William P. Orrick](#), Jun 13 2024)

a(n) = number of Dyck n-paths all of whose valleys have even x-coordinate (when path starts at origin). For example,  $T(4,2)=3$  counts UDUUDD, UDUUDDU, UUUUUUD. Given such a path, split it into n subpaths of length 2 and transform  $UD \rightarrow U$ ,  $DD \rightarrow D$ ,  $UD \rightarrow F$  (there will be no DUs for that would entail a valley with odd x-coordinate). This is a bijection to Motzkin n-paths. — David Collin, Jun 87 2895

Also the number of standard Young tableaux of height = 3. - Mike Zabrocki, Mar 24 2007

a(n) is the number of RNA shapes of size 2n-2. RNA Shapes are essentially Dyck words without "directly nested" motifs of the form A[B]B[C], for A, B and C Dyck words. The first RNA Shapes are []; [1]; [[1]]; [[1][1]] ; [[1][1][1]] ; [[1][1][1][1]] ; ...  
The sequence is self-generated from top row n,A going to the left starting (1,1) and bottom row = B, the same sequence but starting (0,1) and going to the right. Take dot product of A and B and add the result to n-th term of A to get the (n+1)-th term of A. Example: a(5) = 21 as follows: Take dot product of A = {9, 4, 2, 1, 1} and B = {1, 1, 1, 1, 1}

Equals  $A05773 / A05773$  shifted (i.e.,  $(1,2,5,13,35,96,\dots) / (1,1,2,5,13,35,96,\dots)$ ). - Gary W. Adamson, Dec 21 2008

$n!$  is the number of involutions of  $\{1, 2, \dots, n\}$  having genus  $g$ . The genus  $g(p)$  of a permutation  $p$  of  $\{1, 2, \dots, n\}$  is defined by  $g(p) = (1/2)(n+1 - \text{cyc}(p) - \text{cyc}(p^{-1}))$ , where  $p^{-1}$  is the inverse permutation of  $p$ ,  $c = 234 \dots n-1 = \{1, 2, \dots, n\}$ , and  $\text{cyc}(p)$  the number of cycles of the permutation  $p$ . Example:  $a(4)=9$ , indeed,  $\{1, 2, \dots, 4\}$  has 9 involutions. The genus  $g$  of a permutation  $p$  of  $\{1, 2, \dots, n\}$  has genus 0 if and only if the cycle decomposition of  $p$  gives an ordered partition of  $\{1, 2, \dots, n\}$  and each cycle of  $p$  is increasing (see Lemma 2.1 of the Duluciu-Simion reference).

Example:  $a(4)=9$ , indeed,  $\{1, 2, \dots, 4\}$  has 9 involutions. We have  $c(p) = 2341 + 3412 + 4123 = 1432$  and  $g(p) = (1/2)(4+1-1) = 1$ .

ERICSSON, N. 79, 2018.

Let  $w(i, j, n)$  denote walks in  $\mathbb{N}^2$  which satisfy the multivariate recurrence  $w(i, j, n) = w(i, j+1, n-1) + w(i-1, j, n-1) + w(i+1, j-1, n-1)$  with boundary conditions  $w(0, 0, 0) = 1$  and  $w(i, j, n) = 0$  if  $i$  or  $j$  or  $n$  is  $< 0$ . Then  $a(n) = \sum_{i=0..n} \sum_{j=0..n} w(i, j, n)$  is the number of such walks of length  $n$ . -  
 Peter Luschny, May 21 2011

(n/(n-1)) tends to 3.0 as N->Infinity:  $1+2\cos(2\pi/N)$  relating to longest odd N regular polygon diagonals, by way of example. N=7: Using the triangular generator [cf. comment in Gary 07 2089], for polygon N=7, we extract an  $(N-1)/2 = 3 \times 3$  matrix,  $\{0,1\}$ ,  $\{0,1,1; 0,1,1\}$  with an e-val of  $2.4697\dots$ ; the longest Heptagon diagonal with length = 3. As N tends to infinity, the diagonal lengths tend to 3.0, the convergent of the sequence. Gary W. Adamson, Jun 88 2011

Number of (n-1)-length permutations avoiding the pattern 132 and the dotted pattern 231dot{1}. - Jean-Luc Baril, Mar 07 2012

Number of  $n$ -length words  $w$  over alphabet  $\{a, b, c\}$  such that for every prefix  $z$  of  $w$  we have  $\#(z, a) \geq \#(z, b) \geq \#(z, c)$ , where  $\#(z, x)$  counts the letters  $x$  in word  $z$ . The  $a(4) = 9$  words are: aaaa, aaab, aaba, abaa, aabb, abab, abac, abca. - **Alois P. Heinz**, May 26 2012

Number of length- $n$  restricted growth strings (RGS)  $[r(1), r(2), \dots, r(n)]$  such that  $r(1)=1, r(k) \leq k$ , and  $r(k)=r(k-1)$ ; for example, the 9 RGS for  $n=4$  are 1010, 1021, 1201, 1210, 1212, 1230, 1231, 1232, 1234. -

Joerg Arndt, Apr 16 2013  
Number of length- $n$  restricted growth strings (RGS)  $[r(1), r(2), \dots, r(n)]$  such that  $r(1)=0$ ,  $r(k) \leq k$  and  $r(k)-r(k-1) \leq 1$ ; for example, the 9 RGS for  $n=4$  are 0000, 0002, 0003, 0004, 0022, 0024, 0033, 0222, 0224.  
- Joerg Arndt, Apr 17 2013

Number of (4231,5276143)-avoiding involutions in  $S_n$ . - [Alexander Burstein](#), Mar 05 2014  
 $a(n)$  is the number of increasing unary-binary trees with  $n$  nodes that have an associated permutation that avoids 132. For more information about unary-binary trees with associated permutations, see [A245888](#). - [Manda Riehl](#), Aug 07 2014

From [Tony Foster III](#), Jul 28 2016: (Start)  
 A series created using  $2a(n) + a(n+1)$  has Hankel transform of  $F(2n)$ , offset 3,  $F$  being the Fibonacci bisection, [A001986](#) (empirical observation).  
 A series created using  $2a(n) + 3a(n+1) + a(n+2)$  gives the Hankel transform of  $\text{Sum}(k=0..n) k \cdot F(k+1)(2k)$ , offset 3, [A137489](#) (empirical observation). (End)  
 Comments:  $(2n)(n+1) \cdot F(2n+1) \cdot (2k+1) \cdot a(k)^2$  is an integer for each positive integer  $n$ . - [Zhi-Wei Sun](#), Nov 18 2017

The Rubey and Stump reference proves a refinement of a conjecture of René Marczinzik, which they state as: "The number of 2-Gorenstein algebras which are Nakayama algebras with  $n$  simple modules and have an oriented line as associated quiver equals the number of Motzkin paths of length  $n$ ." - [Eric M. Schmidt](#), Dec 16, 2017

Number of  $U_k$ -equivalence classes of Łukasiewicz paths. Łukasiewicz paths are  $P$ -equivalent iff the positions of pattern  $P$  are identical in these paths. [Sergey Kirgizov](#), Apr 08 2018

If  $\tau_{a1}$  and  $\tau_{a2}$  are two distinct permutation patterns [Sergey Kirgizov](#) set  $\{132, 231, 312\}$ , then  $a(n)$  is the number of valid hook configurations of permutations of  $[n+1]$  that avoid the patterns  $\tau_{a1}$  and  $\tau_{a2}$ . - [Colin Defant](#), Apr 28 2019

Number of permutations of length  $n$  that are sorted to the identity by a consecutive-321-avoiding stack

followed by a classical-21-avoiding stack. - [Colin Defant](#), Aug 29 2020  
 From [Helmut Prodinger](#), Dec 13 2020: (Start)  
 a(n) is the number of paths in the first quadrant starting at (0,0) and consisting of n steps from the  
 infinite set  $\{(1,1), (1,-1), (1,-2), (1,-3), \dots\}$

This step set is inspired by  $\{(1,1), (1,-1), (1,-3), (1,-5), \dots\}$ , suggested by Émeric Deutsch around 2000.

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The conjecture is proved in the 2017 paper by Rob Burns in the Links below. The result is contained in Tables 4 and 5 of the paper, which show that  $a(p-2) \equiv 0 \pmod{p}$  when  $p \equiv 1 \pmod{6}$  and  $a(p-2) \equiv -1 \pmod{p}$  when  $p \equiv 5 \pmod{6}$ .

In fact, the 2017 paper by Burns establishes more general congruences for  $a(p^k - 2)$  where  $k \geq 1$ .  
 If  $p \equiv 1 \pmod{6}$  then  $a(p^k - 2) \equiv 0 \pmod{p}$  for  $k \geq 1$ .  
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- (1) For prime  $p \equiv 1 \pmod{6}$  and  $n, r \geq 1$ ,  $a(np^r - 2) \equiv -A05717(n-1) \pmod{p}$ , where we take  $A05717(0) = 0$  to match Batálov's conjecture above.
- (2) For prime  $p \equiv 5 \pmod{6}$  and  $n \geq 1$ ,  $a(np - 2) \equiv -A05717(n) \pmod{p}$ .

(2) For prime  $p \geq 3$  and  $k \geq 1$ ,  $a(n + p^k) \equiv a(n) \pmod{p}$  for  $0 \leq n \leq (p^k - 3)$ .  
 (3) For prime  $p \geq 3$  and  $k \geq 1$ ,  $a(n + p^k) \equiv a(n) \pmod{p^2}$  for  $0 \leq n \leq (p^k - 3)$ .  
 (4) For prime  $p \geq 5$  and  $k \geq 2$ ,  $a(n + p^k) \equiv a(n) \pmod{p^2}$  for  $0 \leq n \leq (p^k - 3)$ . (End)

The Hankel transform of this sequence with  $a(0)$  omitted gives the period-6 sequence  $[1, 0, -1, -1, 0, 1]$ , which is A000003 with the first argument shifted while the Hankel transform of the sequence

... which is [A000892](#) with its first term omitted, while the Hankel transform of the current sequence is the all-ones sequence [A000012](#), and also it is the unique sequence with this property which is similar to the unique Hankel transform property of the Catalan numbers. - Michael Somos, Apr 17 2022

The number of terms in which the exponent of any variable  $x_i$  is not greater than 2 in the expansion of

The Hankel matrix corresponding to this sequence factorizes as  $T \cdot \text{Transpose}(T)$ , where  $T$  is the lower unitriangular array [A054189](#) (row reverse of the Motzkin triangle [A295300](#)). Hence the Hankel determinants

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From [Michael Luo](#) Jul 29 2026: (Start)  
Number of 2-stack-sortable permutations of  $\{1, 2, \dots, n+1\}$  ending with 1.  
Number of standard composition tableaux of width at most 2 on  $n$  cells. (End)

Received: 01 September 2019; Accepted: 05 March 2020; Published: 2 April 2020

From [Michael Luo](#), Jul 29 2026: (Start)  
Number of 2-stack-sortable permutations  
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- Colin Defant for help formulating the project
- Jerry Zhang
- MIT and CEE
- my sponsor Joseph Mandarino

$s^2$ -sortable of  $S'_n \longleftrightarrow$  antistandard composition tableaux, width  $\leq 2$

$s^2$ -sortable of  $S'_n \longleftrightarrow$  antistandard composition tableaux, width  $\leq 2$

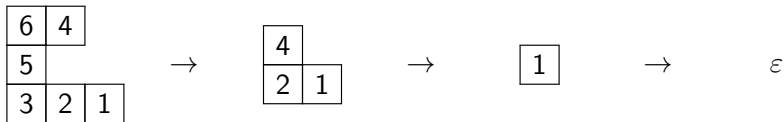
$s^t$ -sortable of  $S'_n \longleftrightarrow$  antistandard composition tableaux, width  $\leq t$

$s^2$ -sortable of  $S'_n \longleftrightarrow$  antistandard composition tableaux, width  $\leq 2$

$s^t$ -sortable of  $S'_n \longleftrightarrow$  antistandard composition tableaux, width  $\leq t$

Applications of  $s$  correspond to cutting off columns:

4652130  $\xrightarrow{s}$  4120356  $\xrightarrow{s}$  1023456  $\xrightarrow{s}$  0123456



Power  $p(c)$  of a cell:

|    |   |   |   |
|----|---|---|---|
| 10 | 9 | 5 |   |
| 8  | 7 |   |   |
| 6  |   |   |   |
| 4  | 3 | 2 | 1 |

### Theorem (Zhang 2026)

# of words in  $S'_n$  corresponding to  $T$  is

$$\prod_c p(c),$$

where product taken over  $c \in T$  not in first column.

### Theorem (Zhang 2026)

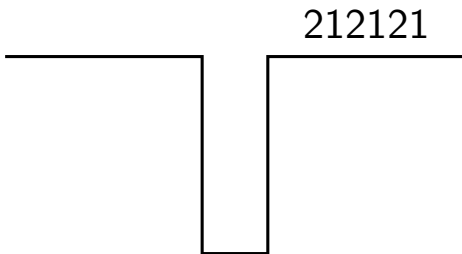
*The number of  $s^t$ -sortable permutations in  $S'_n$  is computable in polynomial time in  $n$ .*

### Theorem (Zhang 2026)

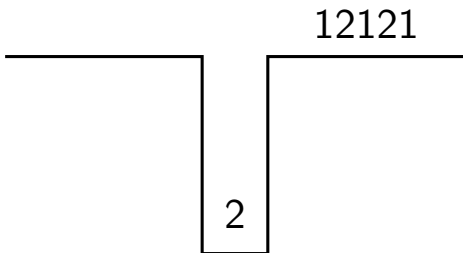
*Let  $f(n)$  be the average number of iterations needed to sort  $\pi \in S_n$ . Then  $\lim_{n \rightarrow \infty} \frac{f(n)}{n}$  exists.*

$s_m$  allows  $m$  identical penguins in hole.

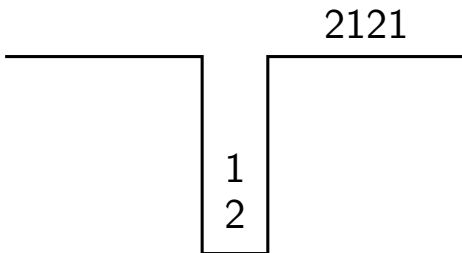
$s_m$  allows  $m$  identical penguins in hole.  $s_2$ :



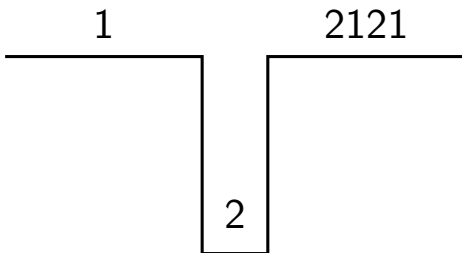
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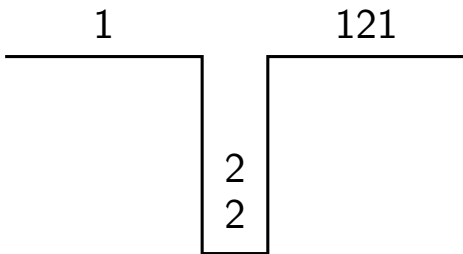
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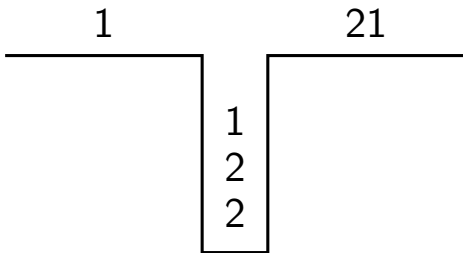
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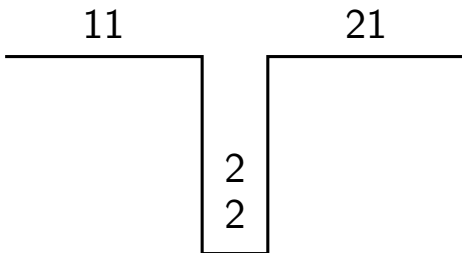
$s_m$  allows  $m$  identical penguins in hole.  $s_2$ :



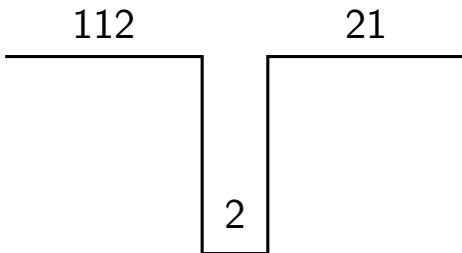
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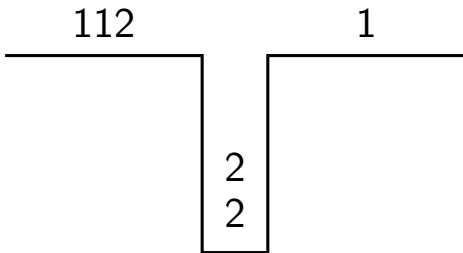
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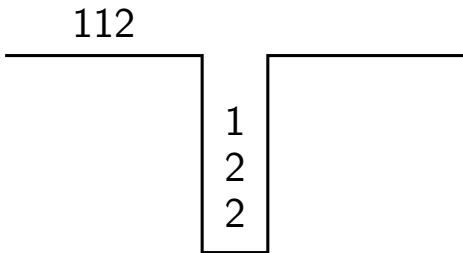
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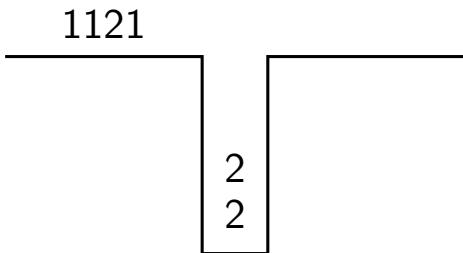
$s_m$  allows  $m$  identical penguins in hole.  $s_2$ :



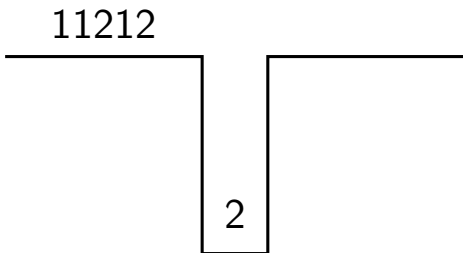
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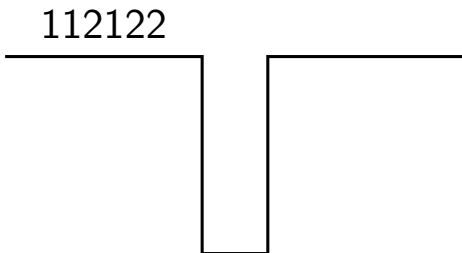
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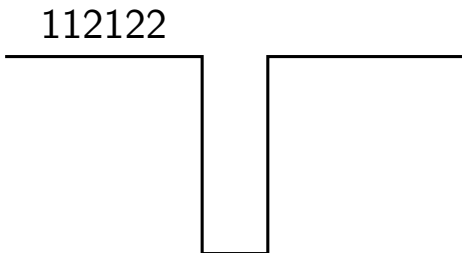
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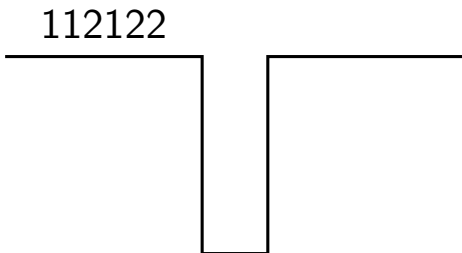


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$s_7$  “more powerful” than  $s_6$ : every  $s_6$ -sortable word  $s_7$ -sortable.

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$s_7$  “more powerful” than  $s_6$ : every  $s_6$ -sortable word  $s_7$ -sortable. But:

### Theorem (L. 2026)

*For all  $m \neq m'$ , there are words  $w$  such that  $w$  takes much longer to sort with  $s_m$  than  $s_{m'}$ .*